

K Nearest Neighbors

Given the following sepal data, classify its species: Length = 5.2, Width = 3.1

Step 1) Find the Euclidean distance from the test point to each point in the sample

Step 2) Rank each point from smallest to largest

Step 3) Use the given k to find the species based on the majority

Length	Width	Species	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Rank	k = 1	k = 3	k = 5
5.3	3.7	Setosa					
5.1	3.8	Setosa					
7.2	3.0	Virginica					
5.4	3.4	Setosa					
5.1	3.3	Setosa					
5.4	3.9	Setosa					
7.4	2.8	Virginica					
6.1	2.8	Versicolor					
7.3	2.9	Virginica					
6.0	2.7	Versicolor					
5.8	2.8	Virginica					
6.3	2.3	Versicolor					
5.1	2.5	Versicolor					
6.3	2.5	Versicolor					
5.5	2.4	Versicolor					

